BIO-RAD

Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, suspension

Species Used Human, HeLa, epithelial carcinoma

Molecules Electroporated DNA: pBLCAT2 series, 4.5 kB

Before the Pulse

Cell Growth Medium MEM + 10% Fetal Calf Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest Log phase

Pre-pulse Incubation 5 minutes on ice

Wash Solution Phosphate Buffered Saline

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 4 °C (from ice to chamber at 25 °C)

Electroporation Medium* Phosphate Buffered Saline

Cuvette Gap 0.4 cm

Cell Density 2 x 10 (6) cells in cuvette

Voltage 0.250 kV

Volume of Cells 0.8 ml

Field Strength 0.625 kV/cm

DNA Concentration 2 to 20 µg

Capacitor 960 µF

DNA Resuspension Buffer TE Buffer (10 mM Tris, 1 mM EDTA)

Resistor (Pulse Controller) Ω none

Volume of DNA 2 to 20 µl

Time Constant 18 to 20 msec

After the Pulse

Outgrowth Medium MEM + 10% Fetal Calf Serum

Outgrowth Temperature 37 °C

Length of Incubation 40 hours

Selection Method or Assay Used CAT assay

Electroporation Efficiency Not determined

Per Cent Survival 20 to 60%

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Nucleic Acid Research, **18**(3): 465-470 (1990).
PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH2PO4, 1.15g Na2HPO4

Name of Submittor Mr. Terence Chong

Institution Address National University of Singapore
Institute Molecular Cell & Biology
Kent Ridge Crecent
Singapore 0511

Survey Number 096